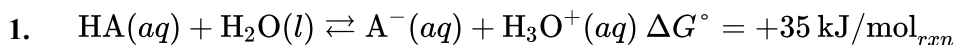
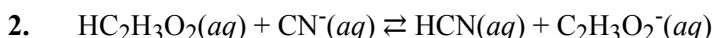


Chapter 11 Part 2: Acidity and Thermo



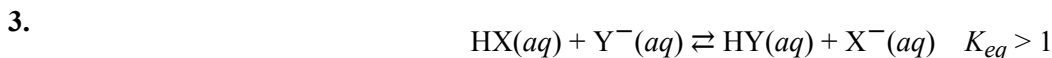
Based on the chemical equation and ΔG° given above, which of the following justifies the claim that $\text{HA}(aq)$ is a weak acid?

- (A) Because $\Delta G^\circ \gg 0$, $K_a \gg 1$, and HA completely dissociates.
- (B) Because $\Delta G^\circ \gg 0$, $K_a \gg 1$, and HA almost completely dissociates.
- (C) Because $\Delta G^\circ \gg 0$, $K_a \ll 1$, and HA only partially dissociates. ✓
- (D) Because $\Delta G^\circ \gg 0$, $K_a \ll 1$, and HA does not dissociate.



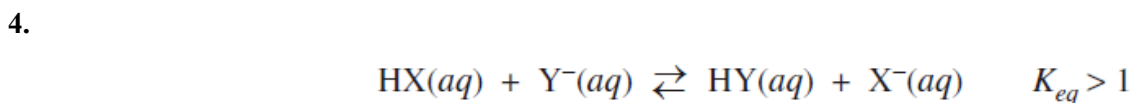
The reaction represented above has an equilibrium constant equal to 3.7×10^4 . Which of the following can be concluded from this information?

- (A) $\text{CN}^-(aq)$ is a stronger base than $\text{C}_2\text{H}_3\text{O}_2^-(aq)$. ✓
- (B) $\text{HCN}(aq)$ is a stronger acid than $\text{HC}_2\text{H}_3\text{O}_2(aq)$.
- (C) The conjugate base of $\text{CN}^-(aq)$ is $\text{C}_2\text{H}_3\text{O}_2^-(aq)$.
- (D) The equilibrium constant will increase with an increase in temperature.
- (E) The pH of a solution containing equimolar amounts of $\text{CN}^-(aq)$ and $\text{HC}_2\text{H}_3\text{O}_2(aq)$ is 7.0.



A solution of a salt of a weak acid HY is added to a solution of another weak acid HX . Based on the information given above, which of the following species is the strongest base?

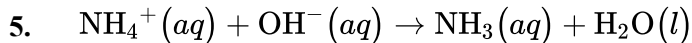
- (A) $\text{HX}(aq)$
- (B) $\text{Y}^-(aq)$ ✓
- (C) $\text{HY}(aq)$
- (D) $\text{X}^-(aq)$



Based on the information given above, which of the following is the strongest acid?

- (A) $\text{HX}(aq)$ ✓
- (B) $\text{Y}^-(aq)$
- (C) $\text{HY}(aq)$
- (D) $\text{X}^-(aq)$

Chapter 11 Part 2: Acidity and Thermo



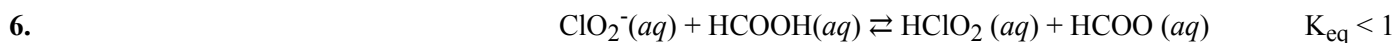
When 0.20 M $\text{NH}_4\text{Cl}(aq)$ and 0.20 M $\text{NaOH}(aq)$ are mixed, the reaction represented by the equation above occurs and a strong smell of ammonia, NH_3 , is observed. Based on this information, which of the following statements is true?

(A) $\text{NH}_4^+(aq)$ acts as a Brønsted-Lowry base.

(B) $\text{NH}_4^+(aq)$ is a stronger acid than $\text{H}_2\text{O}(l)$ is. ✓

(C) $\text{NH}_3(aq)$ is a stronger base than $\text{OH}^-(aq)$ is.

(D) $\text{NH}_4^+(aq)$ and $\text{Cl}^-(aq)$ are a conjugate acid-base pair.



What are the relative strengths of the acids and bases in the reaction represented by the equation above?

(A)	Acid Strength	Base Strength
	$\text{HClO}_2 < \text{HCOOH}$	$\text{ClO}_2^- < \text{HCOO}^-$

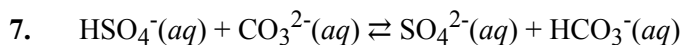
(B)	Acid Strength	Base Strength
	$\text{HClO}_2 < \text{HCOOH}$	$\text{ClO}_2^- > \text{HCOO}^-$

(C)	Acid Strength	Base Strength
	$\text{HClO}_2 > \text{HCOOH}$	$\text{ClO}_2^- > \text{HCOO}^-$

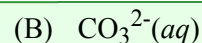
(D)	Acid Strength	Base Strength
	$\text{HClO}_2 > \text{HCOOH}$	$\text{ClO}_2^- < \text{HCOO}^-$

 ✓

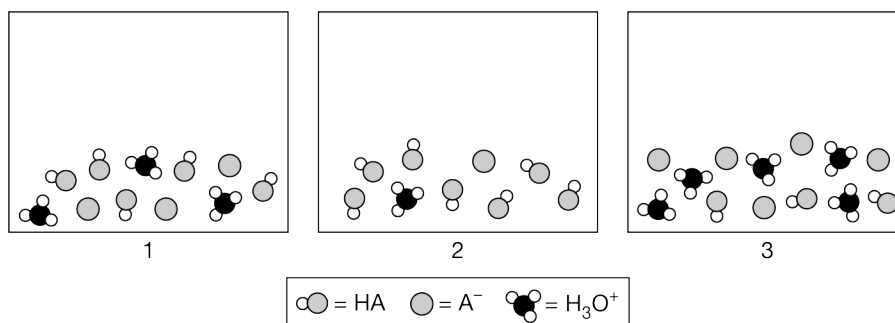
Chapter 11 Part 2: Acidity and Thermo



The reaction represented above is observed to proceed spontaneously to the right in aqueous solution. In this system the strongest base is



8.



Acid
HIO
HBrO
HClO

Each particle diagram shown is a representation of an aqueous solution of one of the acids listed in the table. The molarity of the acids in the solutions is the same. Based on the information, which particle diagram best corresponds to the indicated acid?

(A) Diagram 1 corresponds to HClO.

(B) Diagram 2 corresponds to HClO.

(C) Diagram 2 corresponds to HIO.

(D) Diagram 3 corresponds to HBrO.

Chapter 11 Part 2: Acidity and Thermo

Substance	Structural Formula	Molar Mass (g/mol)
Propylamine	$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{N} - \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array}$	59
Pentane	$\begin{array}{ccccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} - \text{H} \\ & & & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{H} \end{array}$	72
2-propanol	$\begin{array}{ccccc} & & \text{H} & & \\ & & & & \\ & & \text{O} & & \\ & & & & \\ & \text{H} & & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} - \text{H} \\ & & & & \\ & \text{H} & & \text{H} & \end{array}$	60
Propanoic acid	$\begin{array}{ccccc} & \text{H} & & \text{H} & & \text{O} \\ & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} - \text{O} - \text{H} \\ & & & & & \\ & \text{H} & & \text{H} & & \end{array}$	74

A student was given samples of four different unknown liquids. The unknown liquids are propylamine, pentane, 2-propanol, and propanoic acid. The structures and molar masses of the compounds are shown in the table above. On the basis of this information, the student designed some experiments to identify the samples.

9. Of the following, which identifies the aqueous solution of the compound that will have the highest pH and explains why?

(A) Propylamine, because the lone pair of electrons on N can accept an H^+ from the solution



(B) Pentane, because it can produce the largest number of H^+ ions per molecule

(C) 2-propanol, because it can completely dissociate into C_3H_7^+ and OH^-

(D) Propanoic acid, because it dissociates into H^+ and $\text{C}_3\text{H}_5\text{O}_2^-$

Chapter 11 Part 2: Acidity and Thermo

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

Refer to the following compounds.

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (C) CH_3COCH_3
- (D) CH_3COOH
- (E) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$

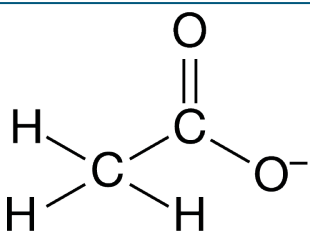
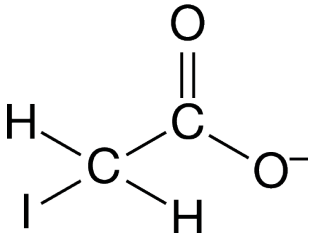
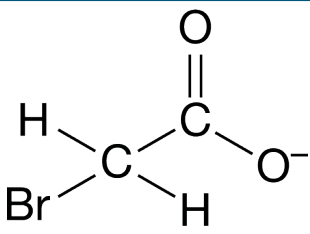
10. Dissolves in water to form an acidic solution

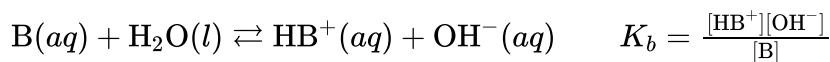
- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (C) CH_3COCH_3
- (D) CH_3COOH
- (E) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$



Chapter 11 Part 2: Acidity and Thermo

11.

Structure	K_b
	5.7×10^{-10}
	1.5×10^{-11}
	7.9×10^{-12}

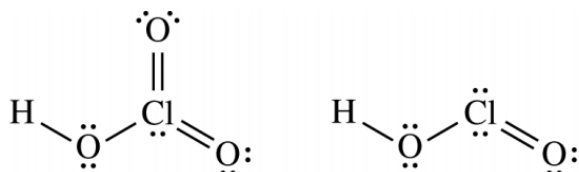


The table above provides the chemical structures for weak bases and their ionization constants, K_b . Based on the data, which of the following provides the best reason for the trend in base strengths?

- (A) The number of hydrogen atoms
- (B) The number of resonance structures
- (C) The different electronegativities of H, I, and Br
- (D) The different molar masses



12.



The Lewis electron-dot diagrams of the HClO_3 molecule and the HClO_2 molecule are shown above at the left and right, respectively. Which of the following statements identifies the stronger acid and correctly identifies a factor that contributes to its being the stronger acid?

Chapter 11 Part 2: Acidity and Thermo

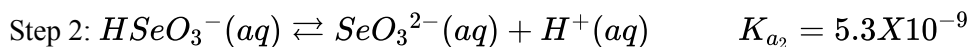
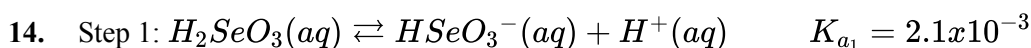
- (A) $\text{HClO}_3(\text{aq})$ is the stronger acid because its molecules experience stronger London dispersion forces.
- (B) $\text{HClO}_3(\text{aq})$ is the stronger acid because the additional electronegative oxygen atom on the chlorine atom stabilizes the conjugate base. ✓
- (C) $\text{HClO}_2(\text{aq})$ is the stronger acid because its molecules experience weaker London dispersion forces.
- (D) $\text{HClO}_2(\text{aq})$ is the stronger acid because the lone pairs of electrons on the chlorine atom stabilize the conjugate base.

13.

Base	CH_3NH_2	NH_2NH_2	HONH_2	ClNH_2
Conjugate Acid	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{N}^+-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{N}-\text{N}^+-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{O}-\text{N}^+-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{Cl}-\text{N}^+-\text{H} \\ \\ \text{H} \end{array}$

The table above includes the Lewis diagrams of the conjugate acids of some weak nitrogenous bases. Based on the relative stability of the conjugate acids, which of the following bases has the largest value for K_b ?

- (A) CH_3NH_2 ✓
- (B) NH_2NH_2
- (C) HONH_2
- (D) ClNH_2



The step-wise dissociation of selenous acid, $\text{H}_2\text{SeO}_3(\text{aq})$, is represented by the equations above. Which of the following best helps explain why the value of K_{a_2} is so much smaller than the value of K_{a_1} ?

- (A) The Se atom in $\text{H}_2\text{SeO}_3(\text{aq})$ is in a higher oxidation state than the Se atom in $\text{HSeO}_3^-(\text{aq})$.
- (B) Water is more attracted to $\text{HSeO}_3^-(\text{aq})$ ions than to SeO_3^{2-} ions, which drives the step 2 equilibrium toward the reactant.
- (C) The $\text{HSeO}_3^-(\text{aq})$ ions produced in step 1 are asymmetrical, but the SeO_3^{2-} ions produced in step 2 are symmetrical.
- (D) Removing the first H^+ from $\text{H}_2\text{SeO}_3(\text{aq})$ requires less energy than removing the second H^+ , because the second H^+ is removed from a negatively charged species. ✓

Chapter 11 Part 2: Acidity and Thermo

15.

Acid	Structure
HClO	$\text{H}-\ddot{\text{O}}-\ddot{\text{Cl}}:$
HBrO	$\text{H}-\ddot{\text{O}}-\ddot{\text{Br}}:$
HIO	$\text{H}-\ddot{\text{O}}-\ddot{\text{I}}:$

A 0.1 *M* aqueous solution of which of the acids shown in the table above would have a pH closest to 7.0, and why?

- (A) HClO, because Cl has the highest electronegativity.
- (B) HBrO, because Br has the lowest electronegativity.
- (C) HIO, because I has the lowest electronegativity. ✓
- (D) All three solutions will have the same pH because they are all monoprotic acids.

16. Which of the following is a weak acid in aqueous solution?

- (A) HCl
- (B) HClO₄
- (C) HNO₃
- (D) H₂S ✓
- (E) H₂SO₄



Which of the following species is the strongest base in the reaction represented above?

- (A) NH₃(aq)
- (B) CH₃NH₃⁺(aq)
- (C) NH₄⁺(aq)
- (D) CH₃NH₂(aq) ✓